Examination –Dec- 2022

B.Tech. I Sem: Common to CSE, CSE BC, IT, AIADS branches

Applied Chemistry

Time :2Hrs

Max. Marks : 60

Min. Marks :19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c &d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part a – Max 50 words, Part b – Max 50 words,

Part c – Max 100 words and Part d – Max 400 words.

Word limit NOT to be followed for diagram, numerical, derivation.

Q.1 (a) Following two questions have four options, out of which only one is correct. Write the correct answer only. 02

(i) Which of the following is *NOT* an Internal Conditioning method?

1. Colloidal Conditioning
2. Calgon conditioning
3. Zeolite conditioning
4. Phosphate conditioning

(ii) Which of the following is a *CORRECT* relationship?

1. 1 ppm = 0.07°Fr
2. 1 °Fr = 1 ppm
3. 1 mg/l = 1 ppm
4. 1 °Cl = 1 ppm

(b) In the Ion exchange resin method of water softening, hard water is first passed through cation exchanger and then through anion exchanger. Why? 02

(c) Name various defects caused by hard water in boiler. Discuss any one of these defects. 03

OR

50 ml of standard hard water containing 1 mg of pure $CaCO_3$ per ml consumed 20 ml of EDTA using Eriochrome Black - T indicator. 03

In another experiment, 100 ml of hard water sample consumed 25 ml of same EDTA solution by same process. Further, 100 ml of same hard water sample after boiling consumed 20 ml of same EDTA solution by same process. Calculate Total, Temporary and Permanent Hardness of water sample.

(d) Discuss Lime soda process of water softening giving only reactions and diagram. 05

OR

What is alkalinity? Write various ions involved in causing alkalinity with reference to Phenolphthalein and Methyl orange indicators. Write significance of alkalinity. 05

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Q.2 (a) Following two questions have four options, out of which only one is correct. Write the correct answer only. 02

(i) The electrode potential is the tendency of a metal

1. To gain electrons
2. To lose electrons
3. Either to lose or to gain electrons
4. Neither to lose nor to gain electrons

(ii) Which of the following is a rechargeable cell/battery?

1. Alkaline battery
2. Dry cell
3. Lithium ion battery
4. Mercury battery

(b) Why does a dry cell become dead after a long time, even if it has not been used? 02

(c) What are reversible and Irreversible cells? Give examples. 03

OR

Calculate the emf of a cell at 25°C, when the concentration of $ZnSO_4$ and $CuSO_4$ are 0.001 M and 0.1 M respectively. Given: $E^o_{Zn2+} = -0.76$ V and $= E^o_{Cu2+} +0.34$ V. 03

(d) Derive Nernst's equation and give its applications. 05

OR

Discuss construction, working and advantages of Li ion battery. 05

Q.3 (a) Following two questions have four options, out of which only one is correct. Write the correct answer only. 02

(i) The rusting of iron is catalysed by

1. Zinc
2. Acidic environment
3. Vacuum
4. Silver

(ii) In Galvanisation, the layer of following metal is coated over iron

1.Gold
2. Silver
3. Copper
4. Zinc

(b) Rusting of iron is faster in sea water as compared to river water. Why? 02

(c) State and explain Pilling Bedworth rule. 03

OR

Discuss electrochemical or wet corrosion. 03

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(d) Discuss various factors influencing corrosion. 05

OR

Discuss the methods for controlling or preventing corrosion. 05

Q.4 (a) Following two questions have four options, out of which only one is correct. Write the correct answer only. 02

(i) Structural units of polymers are called

1. Thermoplastics
2. Fibers
3. Monomers
4. Resins

(ii) 1 nanometer is equal to

1. 10^{9} cm
2. 10^{-9} cm
3. 10^{9} m
4. 10^{-9} m

(b) LDPE and HDPE differ in density. Why? 02

(c) Give classification of polymers with examples. 03

OR

What is dye sensitized solar cell? Give their applications. 03

(d) What are conducting polymers? Give preparation and applications of any two of following: 05

1.Polyaniline

2.Polypyrrole

3.Polythiophene

OR

Write brief notes on any one of the following: 05

1.Nanomaterials

2.Optical fibers

Q.5 (a) Following two questions have four options, out of which only one is correct. Write the correct answer only. 02

(i) Which is NOT a part of visible region of electromagnetic spectrum

1. Violet light 2. Ultra violet light 3. Yellow light 4. Orange light

(ii) Which is an electroanalytical technique of chemical analysis

1. Conductometry 2. Colorimetry 3. IR spectroscopy 4. UV spectroscopy

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(b) *IR spectroscopy is regarded as Finger Print technique in chemical analysis of organic compounds. Why?* 02

(c) What are electrolytes and non electrolytes? Give examples. 03

OR

Write in brief how pH is determined using pH meter? 03

(d) Discuss principle and instrumentation of Gas Chromatography. 05

OR

Discuss principle and instrumentation of IR spectroscopy. 05

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